Assignment-2

Computer Network BCSC0008

1. Convert the following binary IP address to decimal:

11000000.10101000.00000001.00001010

2. What is the network address of 192.168.5.37 with a subnet mask of 255.255.255.0?

3. What is the purpose of a subnet mask in IP addressing?

4. For IP: **172.16.5.100** with subnet mask **255.255.0.0**, determine the Network ID and Host ID.

5. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

6. Find the broadcast address for IP network 172.16.0.0 with subnet mask 255.255.0.0.

7. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

8. An organization has a class B network and wishes to form subnets for 64 departments. Find the subnet mask?

9. How many hosts can each subnet support if the subnet mask is 255.255.255.192?

10. What is the broadcast address of the subnet that contains the IP address 192.168.1.65/26?

11. Given the network 10.0.0.0/16, how many subnets can be created if you use a /20 subnet mask?

12. If an IP address is 172.16.10.25/20, what is its network address?

13. What is the range of valid host addresses in the subnet 192.168.10.0/27?

14. What is the CIDR notation for a subnet mask of 255.255.255.248?

15. You are given the IP 192.168.100.130/25. Identify the **network address**, **broadcast address**, and **usable host range**.

16. How many total subnets and hosts per subnet are available when a Class B network (255.255.0.0) is subnetted with a /20 prefix?

17. Given the CIDR block 172.16.32.0/19, what are the first and last usable host addresses?

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The forwarding table of a router is shown below.

Subnet Number	*Subnet Mask*	*InterfaceID*
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.224	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID ____.

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A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet Address	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	*E*1
145.36.128.0	/17	*E*2
145.36.64.0	/18	*E*3
145.36.255.0	/24	*E*4
Default	--	*E*5

20. Given the CIDR block **192.168.8.0/21**, divide it into **8 equal subnets**. List the subnet addresses in CIDR notation and indicate the broadcast address of the 5th subnet.